

What Do Sequential Generative Latents Encode?

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Motivation

A **world model** learns a representation it can roll forward to predict the future. DreamerV3 [4] learns one **sequentially** with a recurrent state-space model (RSSM) [3]: each observation becomes a stochastic latent joined to a deterministic recurrent state, trained to reconstruct and to predict its own next latent. A VAE [5] instead learns from **static** frames.

World-model latents are increasingly reused as pretrained representations for control, so their content matters.

What does a sequential generative latent encode beyond a static, per-frame model of the same observations?

Probing [1] reads a target off a frozen representation; ST-DIM [2] did this for Atari state. We ask a **controlled** sequential-vs-static version.

Setup: a controlled comparison

We freeze a DreamerV3 RSSM and a static VAE and probe both. Three fairness controls isolate **sequential structure**:

- same encoder/decoder architecture (VAE trained from scratch);
- same observation model (symlog Gaussian), so likelihoods compare;
- same training frames (the replay buffer).

A probe gap then reflects sequential structure, not architecture or data. The VAE drops recurrence and maps one frame to a Gaussian latent via the (negative) ELBO:

$$\mathcal{L}_\beta(x) = \underbrace{\mathbb{E}_{q(z|x)}[-\log p(x|z)]}_{\text{reconstruction}} + \beta \text{KL}(q(z|x) \parallel \mathcal{N}(0, I)).$$

Task: DMC proprioceptive Walker-Walk [6]; three RSSM seeds (2×500K, 1×1.1M) and one matched VAE each.

Probing battery

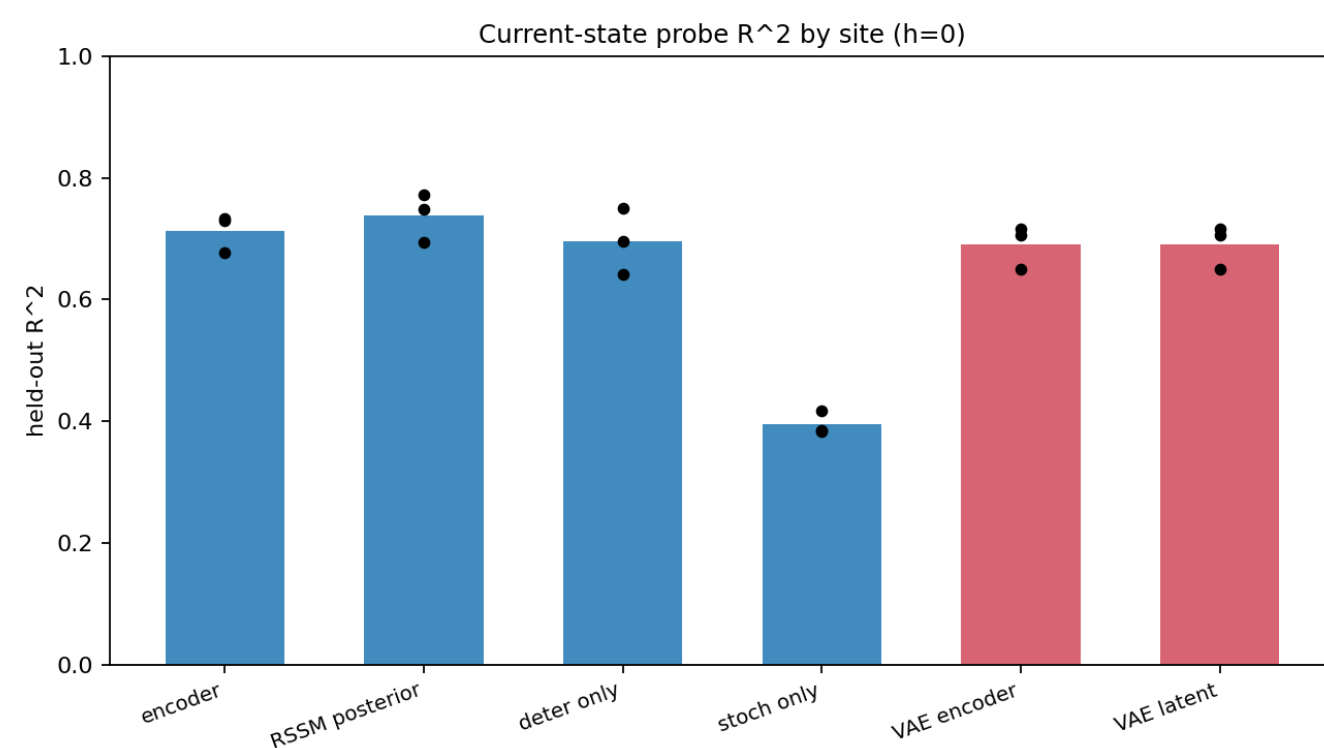
Fit a probe g , report held-out R^2 against the train mean:

$$R^2 = 1 - \frac{\sum (y - g(z))^2}{\sum (y - \bar{y}_{\text{train}})^2}.$$

Same **ridge** (primary) and **MLP** (check) everywhere; episode splits. We vary:

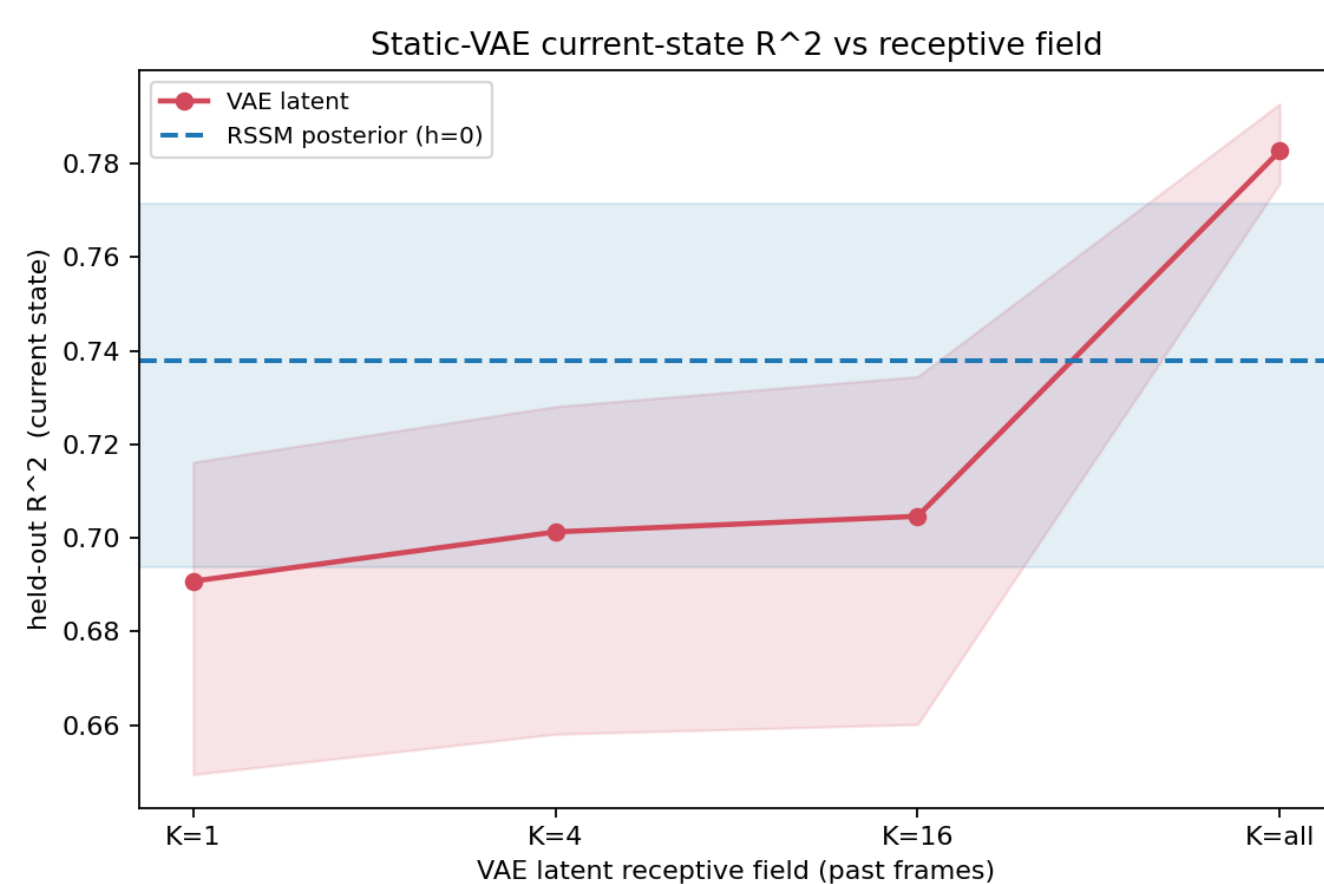
- Targets:** current and future ($k=1, 5, 20$) state; gait phase.
- Sites:** encoder; RSSM posterior (h_t, z_t); open-loop k -step prior; VAE latent.
- Receptive field:** VAE with $K=1, 4, 16$, all causal frames.
- Baseline:** learned VAE-latent dynamics $g(z_t, a_t) \approx z_{t+1} - z_t$, the static analog of the prior.

Result 1: per-frame current state is shared



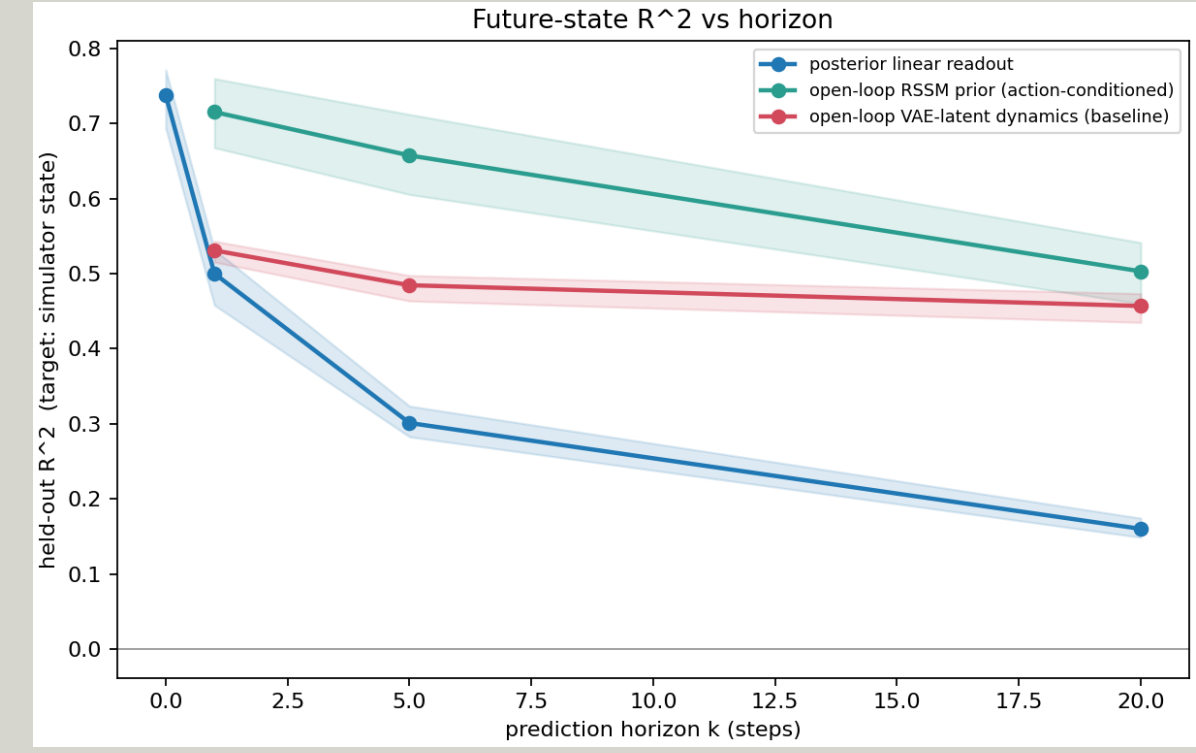
Encoder, RSSM posterior, RSSM deterministic state, and VAE all decode current state at $R^2 \approx 0.69$ – 0.74 . Within the RSSM it sits mainly in the **deterministic state** (0.70); the stochastic latent alone is weaker (0.40 linear, 0.55 MLP). At equal width the continuous VAE latent (0.69) is the stronger linear code.

Result 2: static pooling closes the gap



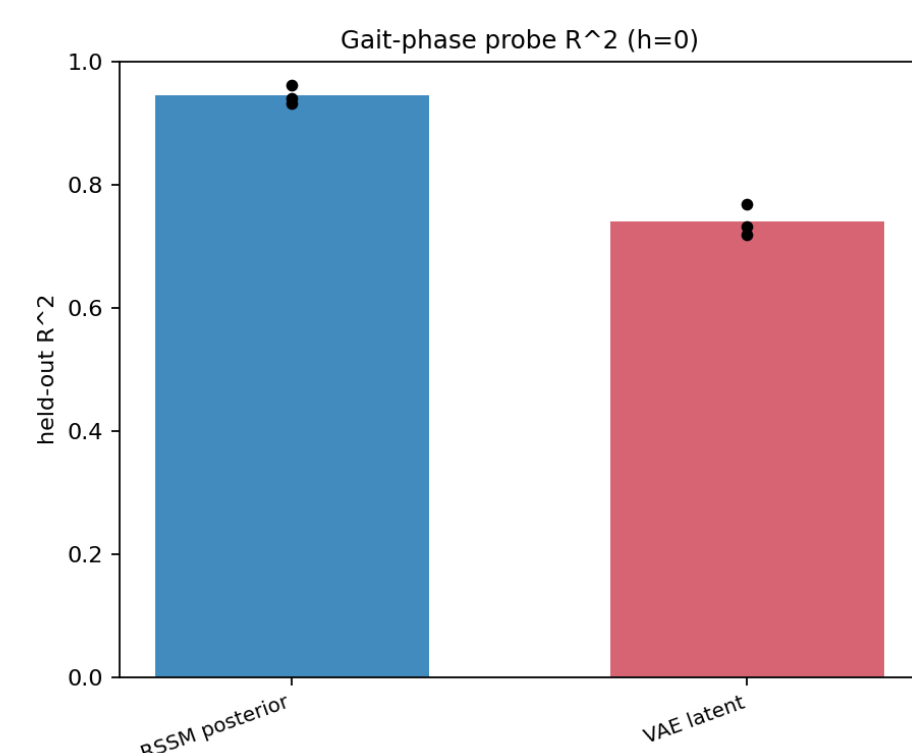
VAE current-state R^2 rises $0.69 \rightarrow 0.78$ with a causal whole-episode mean, **overtaking the RSSM posterior** (0.74). A few recent frames ($K=4, 16$) are not enough. So this content is recoverable by hand-specified pooling, not learned recurrence.

Result 3: prediction lives in the RSSM prior



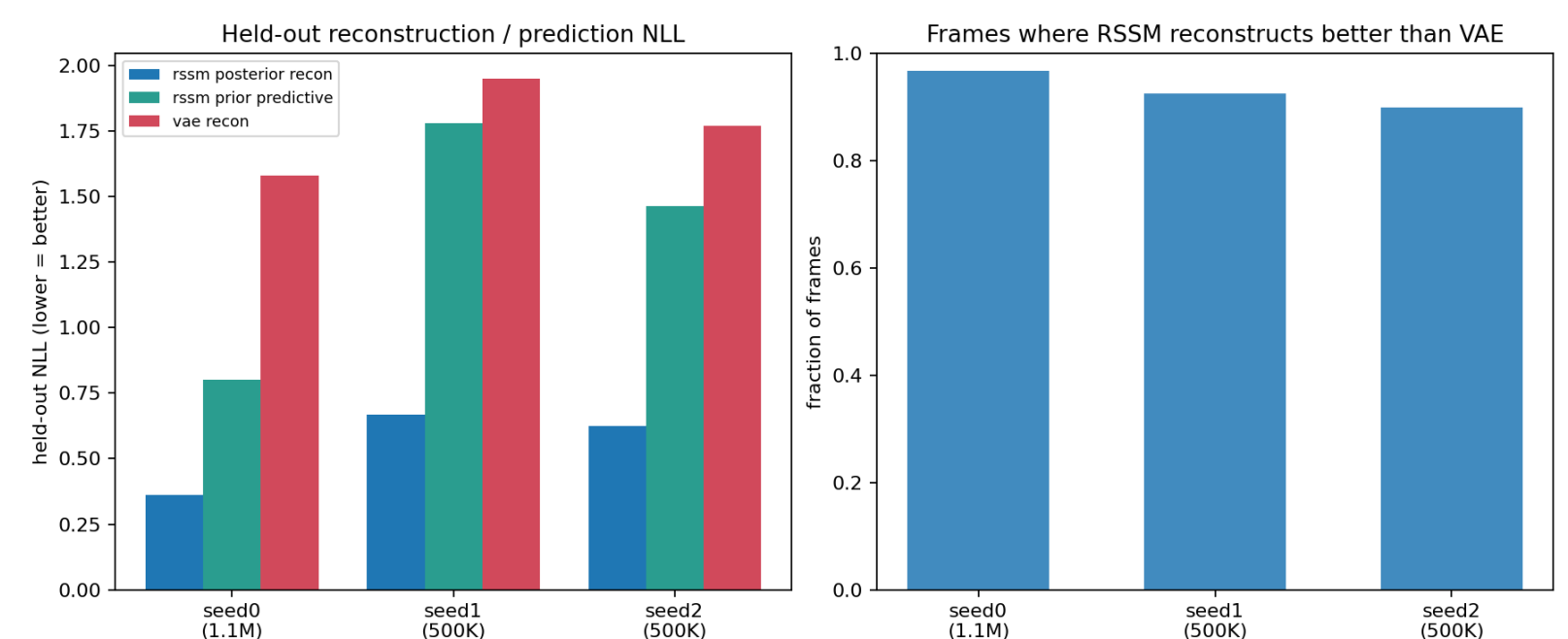
Rolling the **prior** forward retains higher future-state R^2 than a linear readout of the posterior: at $k=20$, $R^2 \approx 0.50$ vs 0.16 (about 3×). A learned VAE-latent dynamics model is a **strong baseline** (0.46). The prior still leads within every seed under the linear probe, but the margin shrinks (+0.18 at $k=1$ to +0.05 at $k=20$) as the horizon grows.

Result 4: gait phase becomes linearly decodable



Largest linear gap: gait phase decodes at $R^2 \approx 0.95$ (posterior) vs 0.74 (VAE), +0.21. A single frame cannot place position within a cycle. But an **MLP** recovers most of it from the static latent (0.89 vs 0.95): the RSSM makes cyclic structure more linearly accessible, not newly present.

Result 5: likelihood $\neq$ probe quality



The RSSM reconstructs with much lower NLL than the VAE (on 90–97% of frames), the 1.1M seed lower still, yet its current-state probe R^2 barely moves. **Likelihood and linear decodability decouple**.

Conclusions

In near-fully-observed proprioception:

- Current-state decoding: static VAE plus causal pooling is enough.
- RSSM value is **predictive** (the prior) and **coordinate-shaping** (it linearizes gait phase).
- A static representation plus dynamics recovers most predictive content: recurrence is not required to access it.
- Generative likelihood is **not** a proxy for probe quality.

Future work

- Pixel / partially observed** tasks, where recurrence should matter more for current-state inference.
- More seeds and model sizes; a controlled training-length sweep.
- Per-component (not pooled) probing; stronger VAE-dynamics models.
- Does linear accessibility predict **downstream control** transfer?

References

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